

Properties of Logarithms - Nathaniel Juan Putra Winata E

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Student

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Topic: Properties of Logarithms

Status: completed

Lesson Plan

Learning Objectives

- Apply the product, quotient, and power rules to expand and condense logarithmic expressions.
- Solve complex logarithmic equations by applying multiple properties of logarithms simultaneously.
- Critique the validity of logarithmic identity statements through formal algebraic deduction.
- Construct simplified expressions from logarithmic structures with different bases using the change-of-base formula.

Lesson Information

Lesson Sections

1. Spiral Review Warm-up (10 minutes)

Teacher Script:

Hi Nathaniel! Great to see you. Before we dive into new territory, let's sharpen our mental math with some exponent and basic log reviews. Since these are the foundation for today's rules, let's see how fast you can solve these in the chat. Ready? 1: What is the value of log base 2 of 16? 2: Convert 3 to the power of 4 into a log form. 3: Simplify x^5 times x^3 . 4: What is (y^{10}) divided by (y^2) ?

Activities:

- *Chatbox rapid-fire check*
- *Virtual whiteboard exponent vs log conversion*

2. Introduction & Real-World Connection (5 minutes)

Teacher Script:

Excellent work. Today, we are unlocking the 'Cheat Codes' of logs. In fields like data science or acoustics (measuring sound), we often have to multiply massive numbers. Logs let us turn that multiplication into simple addition. It's like turning a complex boss fight into a series of mini-games. By the end of this hour, you'll be able to dismantle any logarithmic expression no matter how messy it looks.

Activities:

- *Discussion on Richter Scale or Decibels*

3. Review of Prior Knowledge (5 minutes)

Teacher Script:

Remember, Nathaniel, $\log_b(x) = y$ means $b^y = x$. Think of logs as searching for the exponent. Since we know how exponents behave (like $x^a * x^b = x^{a+b}$), we can guess that logs might have similar rules. Today we focus on the Product, Quotient, and Power rules. They are the mirror images of our exponent laws.

Activities:

- *Formula matching: Exponent Law to Log Law*

4. Problem Introduction (10 minutes)

Teacher Script:

Let's look at our first tool: The Product Rule. $\log(M) + \log(N) = \log(M*N)$. Notice how adding outside the log becomes multiplication inside. Next, the Quotient Rule: $\log(M) - \log(N) = \log(M/N)$. Finally, the Power Rule, which is my favorite: $\log(M^p) = p * \log(M)$. The exponent just slides down to the front like a playground slide. Let's try to expand $\log(x^2 * y)$. Based on what I just said, which rule do we use first?

Activities:

- *Live demonstration of the three rules*
- *Color-coding parts of the log expression*

5. Guided Practice (15 minutes)

Teacher Script:

Let's get our hands dirty. On the whiteboard, I've written: Expand $\log_2(8x^5 / y)$. This has all three rules! Step 1: Handle the division. Step 2: Split the multiplication. Step 3: Slide those exponents down. Show me your Step 1 in the chat.

Activities:

- *Step-by-step expansion of multi-step log problems*
- *Error Analysis: 'What is wrong with this work?' activity*

6. Independent Practice (15 minutes)

Teacher Script:

You're a natural. Now, I'm setting a timer for 10 minutes. I want you to try the 'Log Maze' on screen. You have to condense three separate logs into one single log to find the exit. If you get stuck, type 'HINT' and I'll jump in. Go for it!

Activities:

- *Timed Condensing Challenge*
- *True/False Validity Test (proving $\log(A+B)$ is NOT $\log A + \log B$)*

7. Consolidation & Reflection (5 minutes)

Teacher Script:

Brilliant work today, Nathaniel. You've mastered the expansion and condensation of logs. Which rule felt the most intuitive? Remember, as you do your homework, look for those hidden exponents. Don't forget to join <https://learn.algonova.id/login> to log your progress for today's session. See you next time!

Activities:

- *Reflection: 'The muddiest point' (what was hardest?)*
- *Exit ticket: Condense $2\log A + 3\log B$*

Homework

Medium Questions:

1. Expand the expression: $\log_b(x^3 \cdot y^2)$
2. Condense the expression: $4 \log_5(m) - \log_5(n)$
3. Evaluate using properties: $\log_2(32) - \log_2(8)$
4. Use the change-of-base formula to approximate $\log_3(20)$ to two decimal places.

Hard Questions:

1. Expand completely: $\log \left[\frac{\sqrt{x^3 \cdot y}}{z^4} \right]$
2. Condense into a single logarithm: $\frac{1}{2} \log(x) + 3 \log(y) - 2 \log(z+1)$
3. Solve for x given: $\log_2(x) + \log_2(x-2) = 3$
4. Prove or disprove: $\log_b(A/B) + \log_b(B/C) + \log_b(C/A) = 0$
5. Expand: $\ln \left[\frac{e^x \cdot x^2}{(x+1)^3} \right]$
6. Simplify to a single number: $\log_4(16^x) / x$

Answer Key:

1. $3\log_b(x) + 2\log_b(y)$
2. $\log_5(m^4 / n)$

3. $\log_2(32/8) = \log_2(4) = 2$
4. $\log(20) / \log(3) \approx 2.73$
5. $\frac{3}{2} \log(x) + \frac{1}{2} \log(y) - 4 \log(z)$
6. $\log \left[\frac{\sqrt{x} \cdot y^3}{(z+1)^2} \right]$
7. $\log_2(x^2 - 2x) = 3 \Rightarrow x^2 - 2x = 8 \Rightarrow (x-4)(x+2)=0$. $x=4$ ($x=-2$ is extraneous).
8. $\log(A/B \cdot B/C \cdot C/A) = \log(1) = 0$. True.
9. $x + 2\ln(x) - 3\ln(x+1)$
10. $\log_4((4^2)^x)/x = \log_4(4^{2x})/x = 2x/x = 2$

Parent Reminder

Today's Math Progress: Logarithmic Laws

Nathaniel tackled Properties of Logarithms today—a challenging Grade 10 topic. He successfully learned how to expand and condense complex expressions using the Product, Quotient, and Power rules. Please encourage him to complete the homework on the portal to solidify these algebraic 'shortcuts'!